Midterm Review

Exercises

Duality

1. Suppose $X \sim Geom(p)$ on $\{1, 2, \dots\}$. Find $P(X > x)$.
2. Suppose $X \sim NegBin(r = 3, p)$ on $\{r, r+1, \dots\}$. Find $P(X > x)$.
3. Draw cards until the 1st ace. Let X be the number of cards dealt. Find $P(X > x)$.
4. Draw cards until the 3rd ace. Let X be the number of cards dealt. Find $P(X > x)$.

Regions

5. Suppose $X \sim Geom(p_1)$ on $\{1, 2, 3, \dots\}$, $Y \sim Geom(p_2)$ on $\{1, 2, 3, \dots\}$, and $X \perp Y$. Find:

a) $P(X = Y)$
b) $P(3X < Y)$

6. Suppose $X \sim Pois(\lambda)$, $Y \sim Geom(p)$ on $\{0, 1, 2, \dots\}$, and $X \perp Y$. Find $P(X = Y)$.
7. Suppose $X \sim Bin(n, p_1)$, $Y \sim Geom(p_2)$ on $\{0, 1, 2, \dots\}$, and $X \perp Y$. Find $P(X = Y)$.

Indicators

8. Return $n = 5$ hats randomly back to $n = 5$ people. Let $X = \#$ of matches. Find:

a) $P(X = 5)$
b) $P(X = 4)$
c) $P(X = 3)$
d) $P(X = 0)$

9. Return n hats randomly back to n people. Let $X = \#$ of matches. Find:

a) $P(X = n), P(X = n-1), P(X = n-2)$ for $n \geq 2$
b) $P(X = 0)$ and $P(X = 0)$ asymptotically
c) $E(X)$
d) $Var(X)$

10. Roll a fair die n times. Let $X = \#$ of different faces and $Y = \#$ of missing faces. Thus $k = 6 = X + Y$. Find:

    Example: 2, 2, 2, 1, 5, 3, 3, 1 $(k = 6, n = 8, x = 4, y = 2)$

a) $E(Y)$
b) $Var(Y)$
c) $E(X)$ in terms of $E(Y)$
d) $Var(X)$ in terms of $Var(Y)$

11. Review the following:

$$1 + 2 + \cdots + n = \binom{n+1}{2} = \tfrac{1}{2}n(n+1)$$
$$2(1 + 2 + \cdots + n) = (n+1)_2 = (n+1)n$$

Find each sum.

a) $1 + 2 + \cdots + (n+1)$
b) $1 + 2 + \cdots + n - 1$
c) $2 \cdot [1 + 2 + \cdots + (n-1)]$
d) $2 \cdot [1 + 2 + \cdots + (n-2)]$

12. There are $n = 10$ parking spaces in a line assuming $n \geq 4$. Park $k = 4$ cars and let $X = \#$ of adjacent car pairs. Find:

a) $P(X = 3), P(X = 0), P(X = 1), P(X = 2)$

Examples

	Position									
x	1	2	3	4	5	6	7	8	9	10
$x = 0$	✓			✓		✓				✓
$x = 1$			✓	✓		✓				✓
$x = 2$			✓	✓					✓	✓
$x = 2$			✓	✓	✓					✓
$x = 3$			✓	✓	✓	✓				

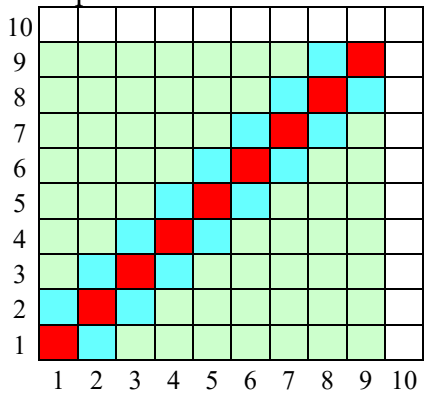
b) $E(X)$
c) $Var(X)$

13. Plant n apple seeds in a line. Each apple seed has a probability of p of growing into an apple tree, independent of other seeds. Let $X = \#$ of adjacent pairs of apple trees. Find:

Example: A _ _ _ A _ **A** A _ A $(n = 10,\ x = 1)$
Example: **A** **A** A _ _ **A** A _ _ _ $(n = 10,\ x = 3)$
Example: **A** **A** **A** **A** A _ _ _ _ _ $(n = 10,\ x = 4)$

a) $E(X)$ in terms of n and p
b) $Var(X)$ in terms of n and p

The I_i's are identical.
The I_i's are not exchangeable.
Example where $n = 10$



$E(I_1) = ?,\ E(I_1 I_2) = ?,\ E(I_1 I_3) = ?$

Notes

Duality

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b) $P(X = 4)$
c) $P(X = 3)$
d) $P(X = 0)$

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b) $P(X = 0)$ and $P(X = 0)$ asymptotically

c) $E(X)$

d) $Var(X)$

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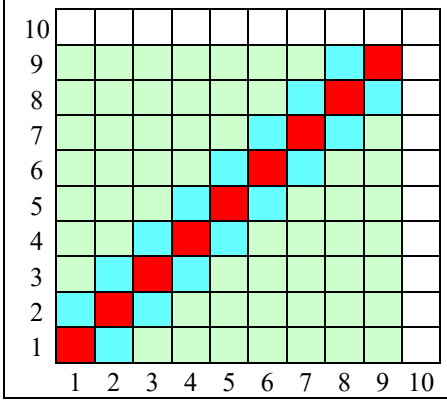
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Examples

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$x = 1$			✓	✓		✓				✓
$x = 2$			✓	✓					✓	✓
$x = 2$			✓	✓	✓					✓
$x = 3$			✓	✓	✓	✓				

b) $E(X)$
c) $Var(X)$

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    Example: A _ _ _ A _ **A** A _ A $(n = 10,\ x = 1)$
    Example: **A** **A** A _ _ **A** A _ _ _ $(n = 10,\ x = 3)$
    Example: **A** **A** **A** **A** A _ _ _ _ _ $(n = 10,\ x = 4)$

The I_i's are identical. The I_i's are not exchangeable. Example where $n = 10$  	$E(I_1) =$?, $E(I_1 I_2) =$?, $E(I_1 I_3) =$?

a) $E(X)$ in terms of n and p
b) $Var(X)$ in terms of n and p